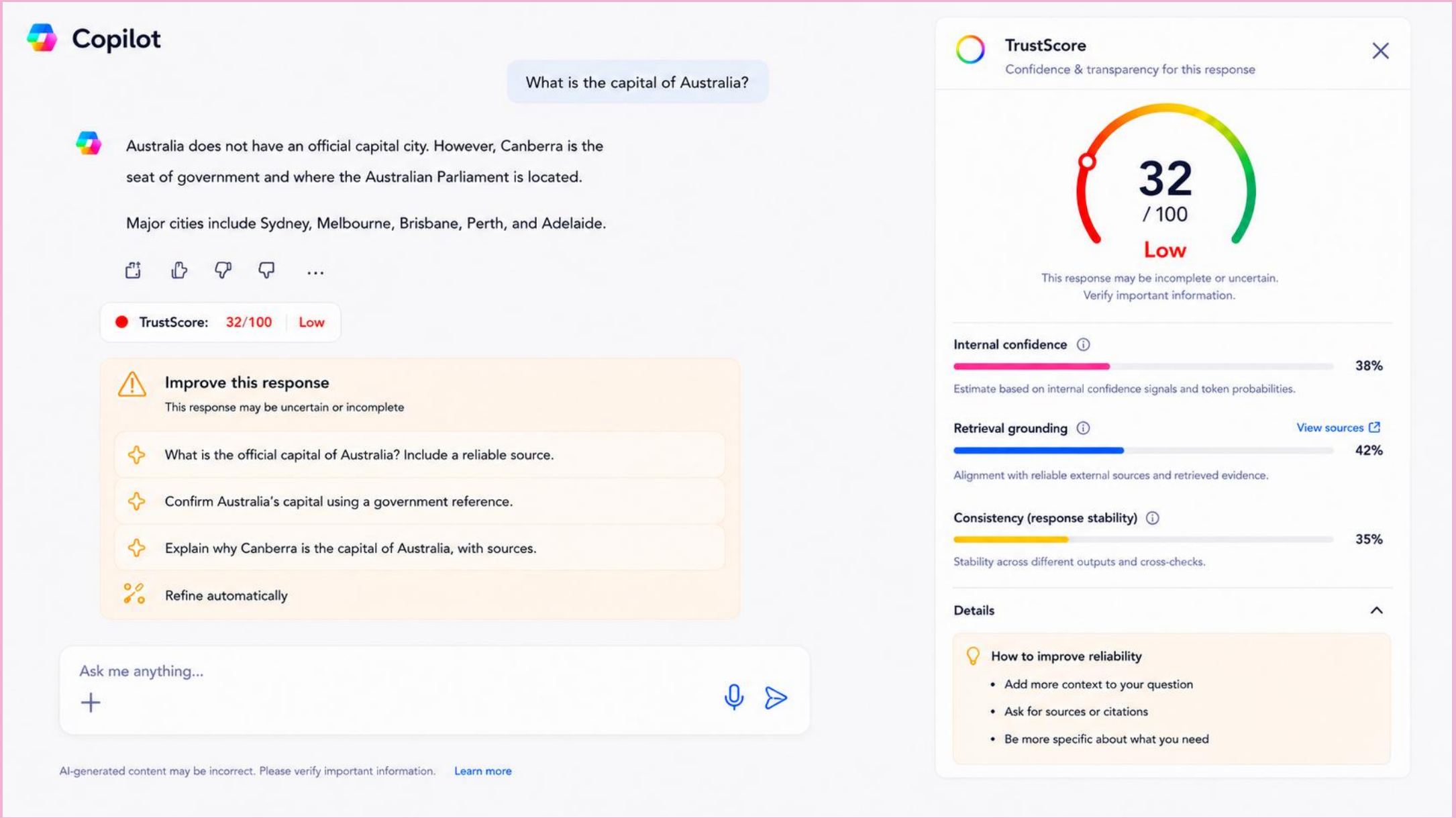
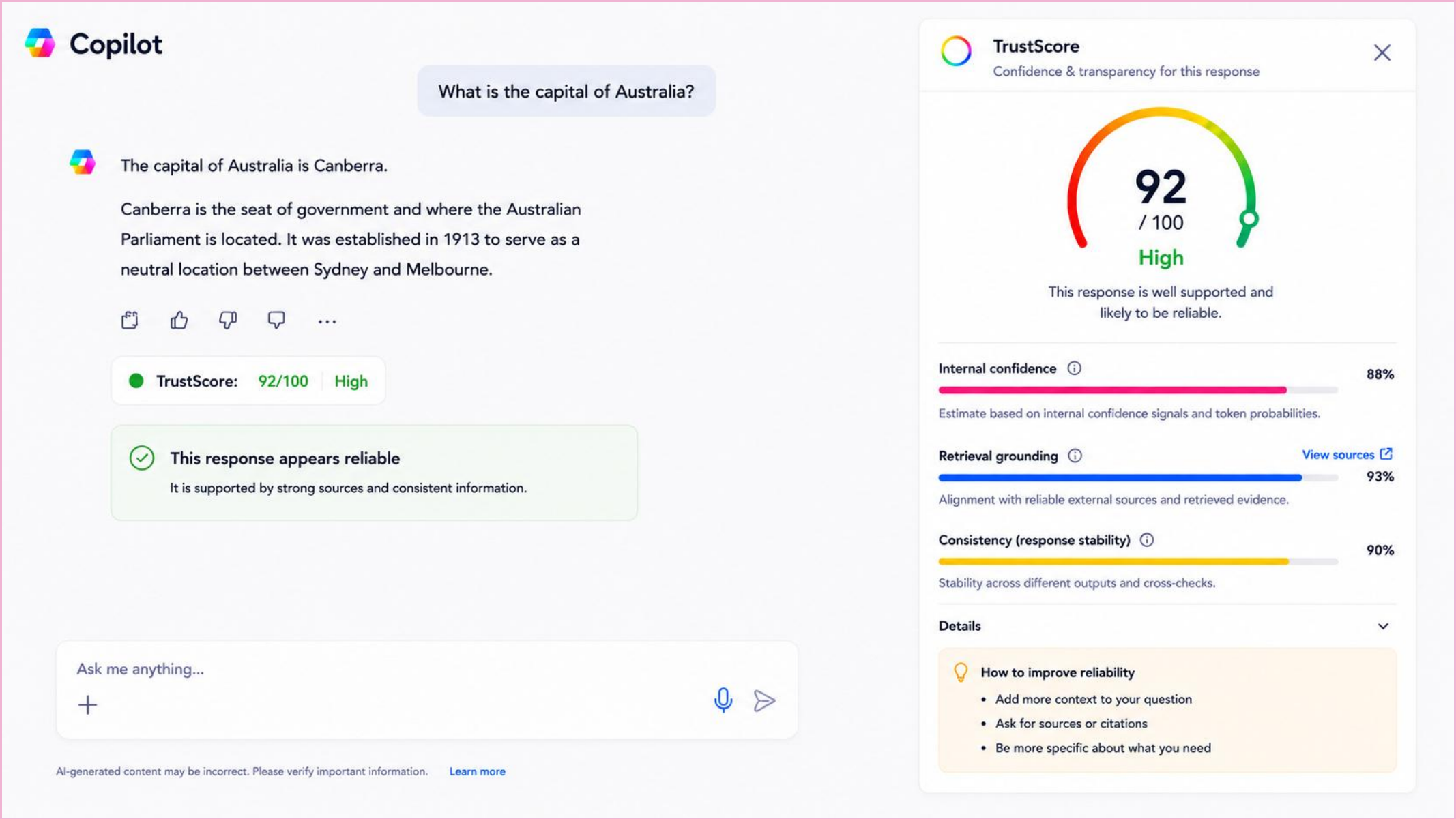
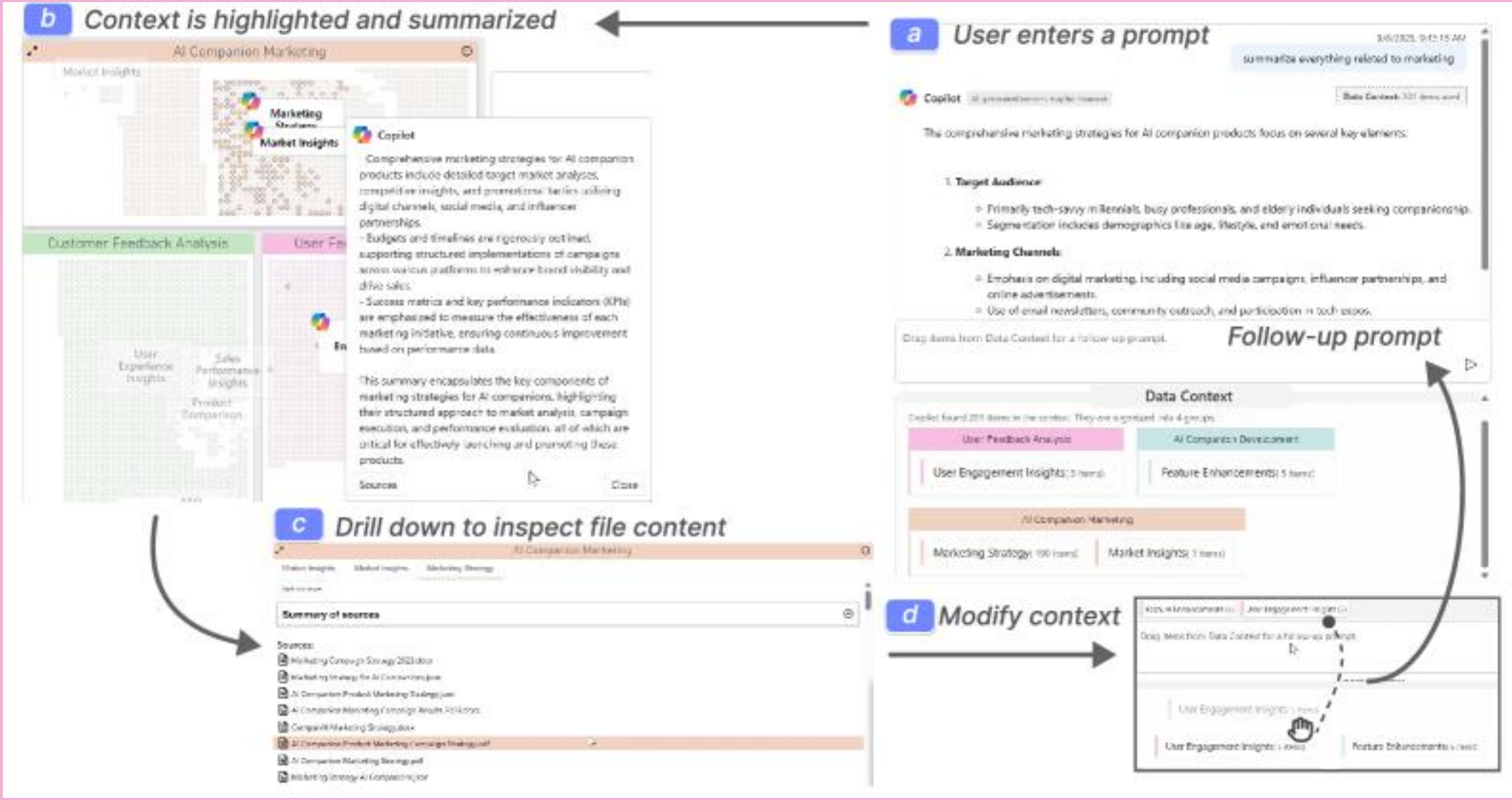


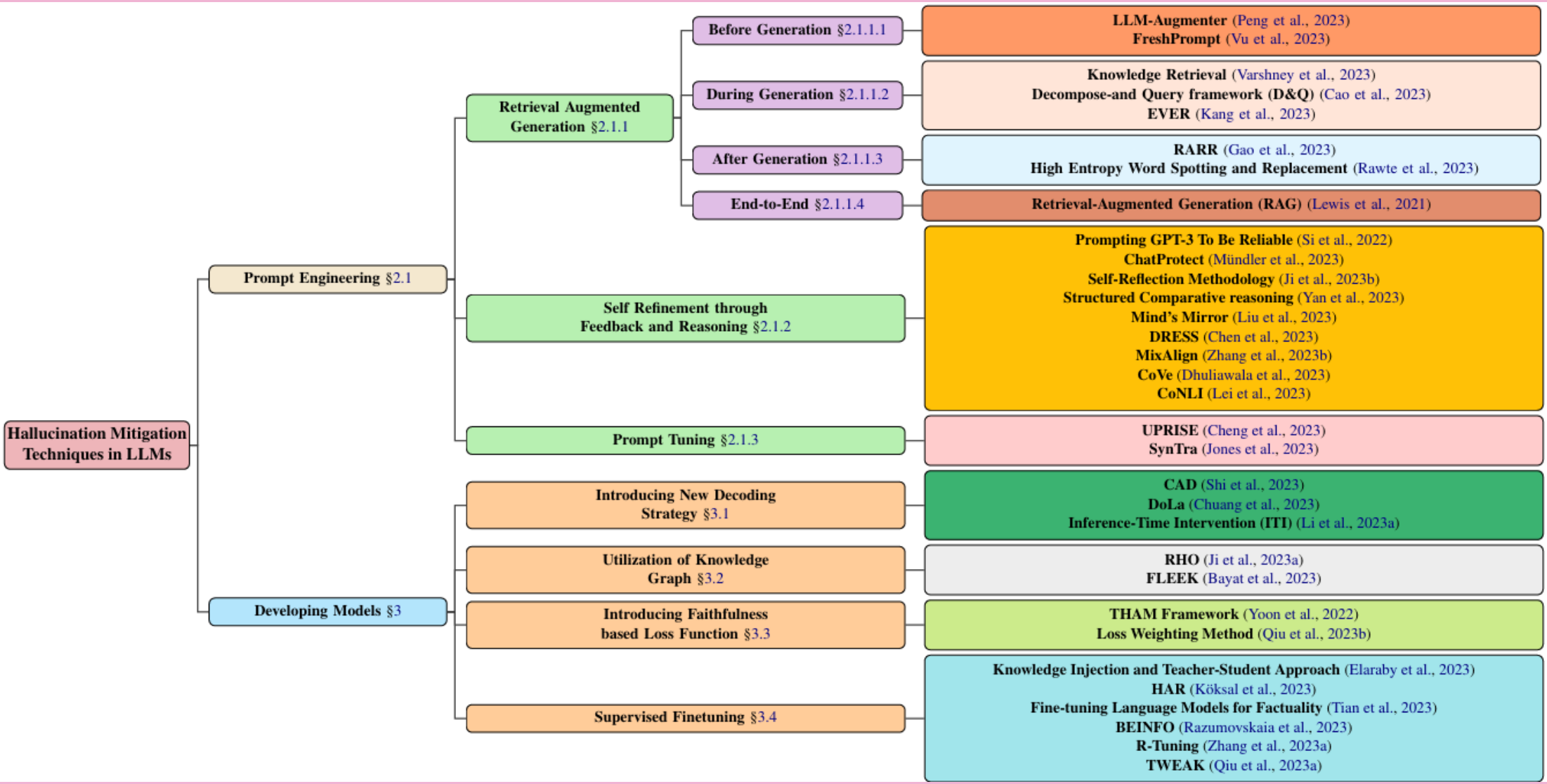
(Source: Open AI, 2026)



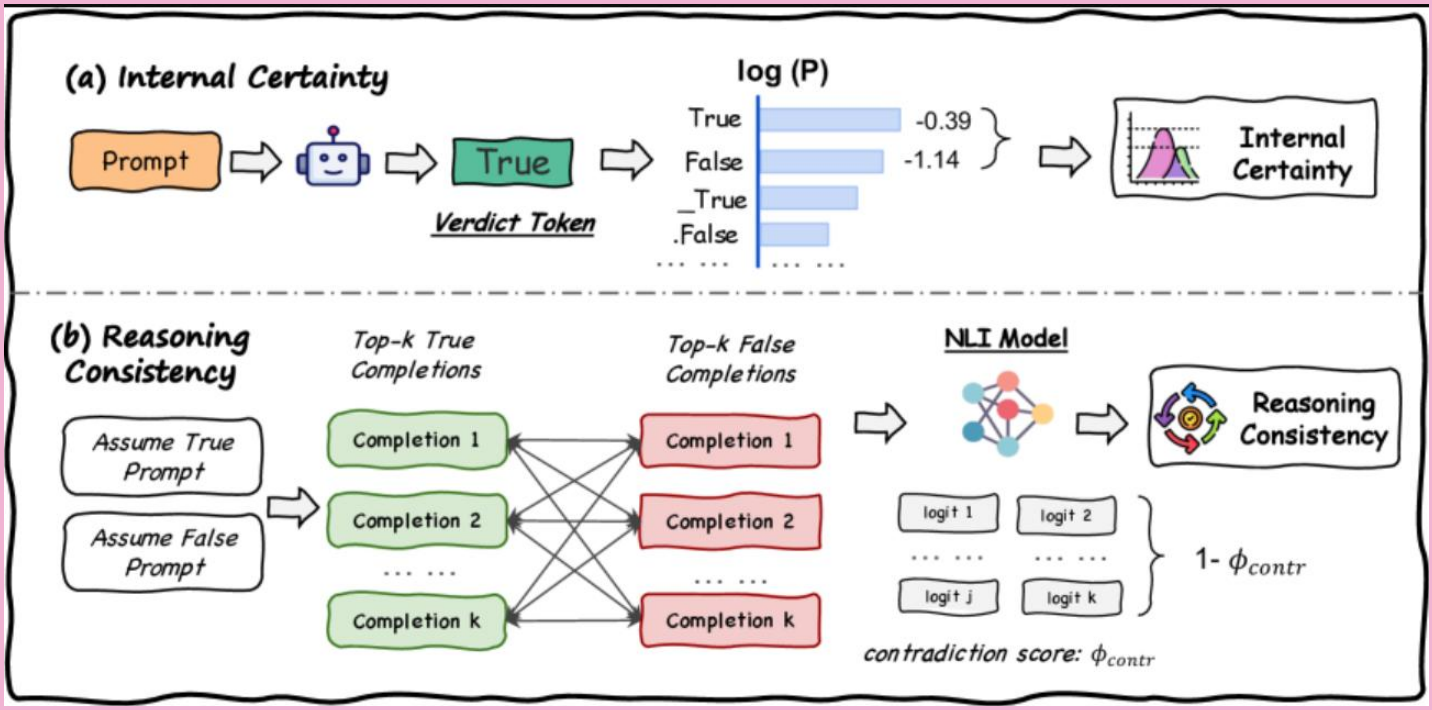
(Source: Open AI, 2026)



(Lee et al., 2025)



(Tonmoy et al., 2025)

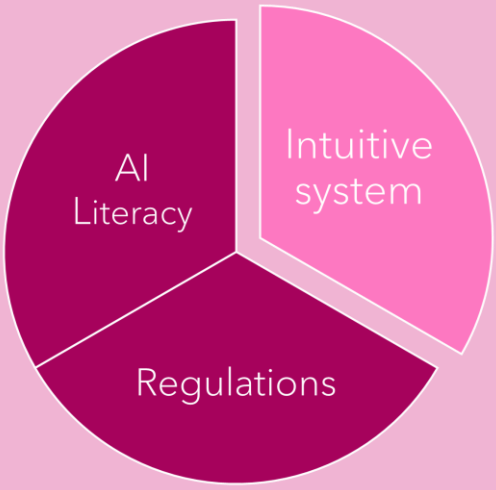


(Wang et al., 2026)

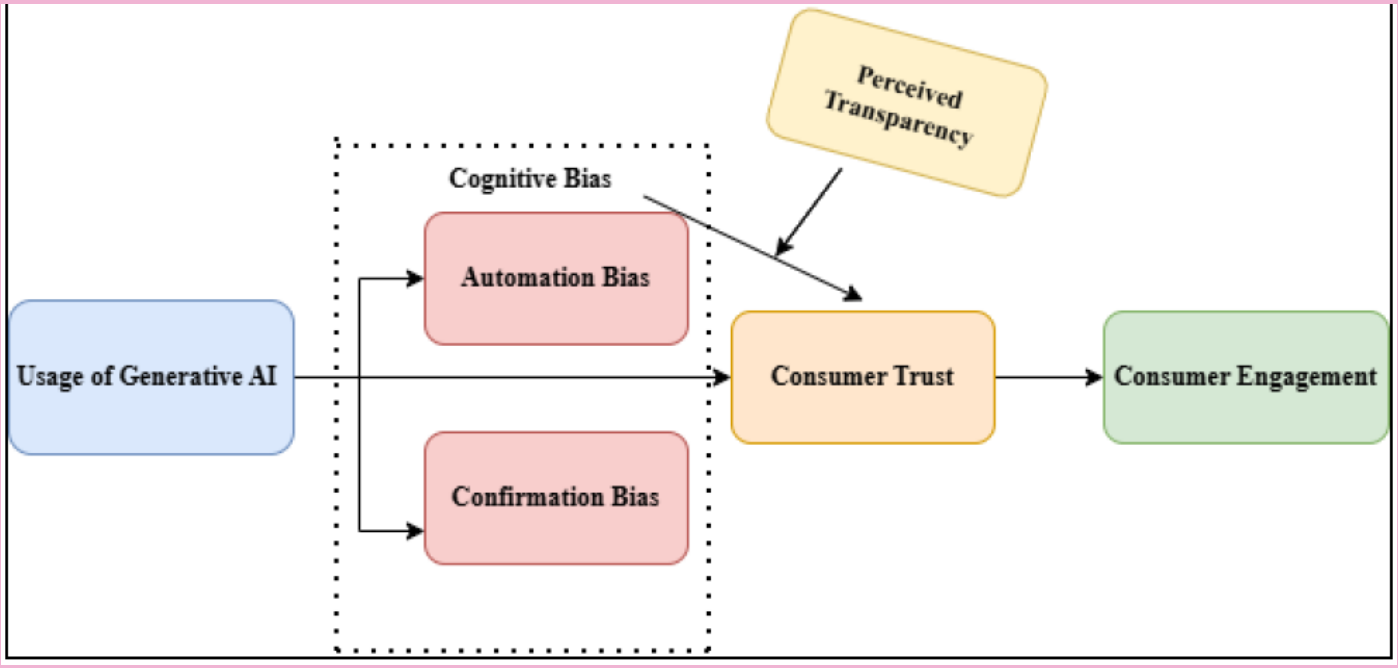
Reasoning consistency & internal certainty (Wang et al., 2026)	
High certainty + high consistency → Direct answering – Factually reliable	Low certainty + low consistency → Deep search – High risk
High certainty + low consistency → Targeted search via reasoning consistency signals – Hallucination, unreliable	Low certainty + high consistency → Targeted search via self-reflection – Knowledge gap lacking factual content

Recommendations

1. Develop a real-time TrustScore UI
    - Implement confidence indicators to communicate the estimated reliability of AI-generated responses in real-time through intuitive visual cues.
2. Build a three-layer confidence engine on top of existing AI models
    - Develop a confidence and transparency layer that evaluates responses through token probability (internal certainty), source grounding (RAG), and cross-output consistency, without retraining underlying LLM models.
3. Integrate user-verification for low-confidence outputs
    - When confidence falls below a threshold, alert users and encourage verification or prompt refinement.







(Ijaz et al., 2025)

- Despite contradictory evidence, users misplace trust in sophisticated AI responses (Tomitza et al., 2025). Requirement analysis indicates that human-centered transparency systems, including intuitive uncertainty indicators, source verification labels and guidance across AI literacy levels, are more effective than prompt engineering, education or legal regulations alone.
- The PCC framework demonstrates that combining probabilistic certainty and reasoning consistency outperforms verbal confidence and fact-checking baselines across LLMs, improving false claim detection by up to 15.2% (ECE), through adaptive verification (Wang et al., 2026).
- A taxonomy of hallucination mitigation techniques, including retrieval grounding, verification systems, and uncertainty estimation confirms a single method is insufficient. A layered approach is required. (Tonmoy et al., 2024).
- RAG reduces hallucinations through external grounding, but may still fail when retrieved information is incomplete, irrelevant or contradictory (Tonmoy et al., 2024; Lee et al., 2025). VizCopilot demonstrates that transparency and verification layers can be implemented on top of existing LLM infrastructure without retraining the underlying model.
- Well-calibrated uncertainty information improves human decision accuracy and confidence calibration compared to AI predictions alone, with visualisation style having minimal effect on outcomes (Marusich et al., 2024).